

Maternal Mental Health and Infant Sleep Analysis

Professional Data Analyst Portfolio Report

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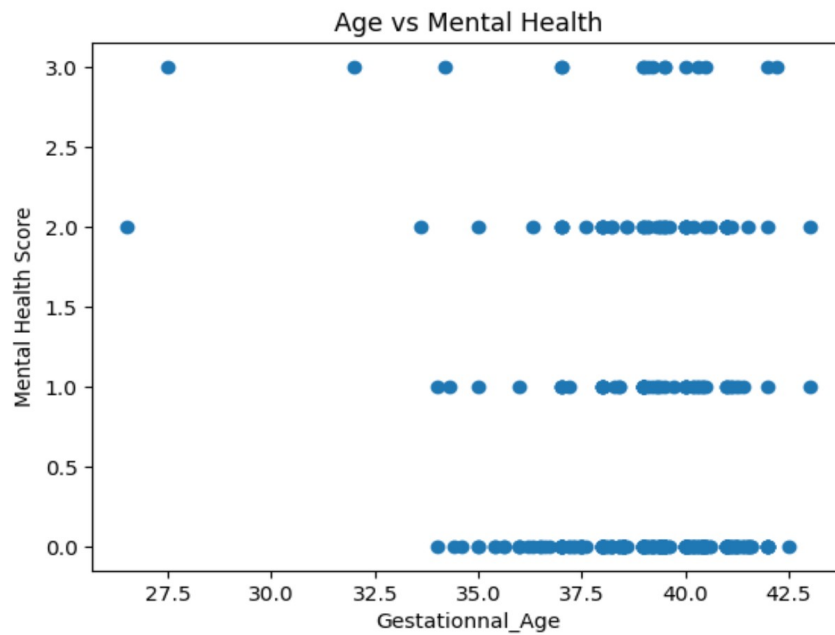
Executive Summary

This project explores relationships between maternal depression (EPDS), anxiety (HADS), childbirth trauma (CBTS), infant sleep patterns, education level, baby age, and baby sex using a dataset of 410 mothers.

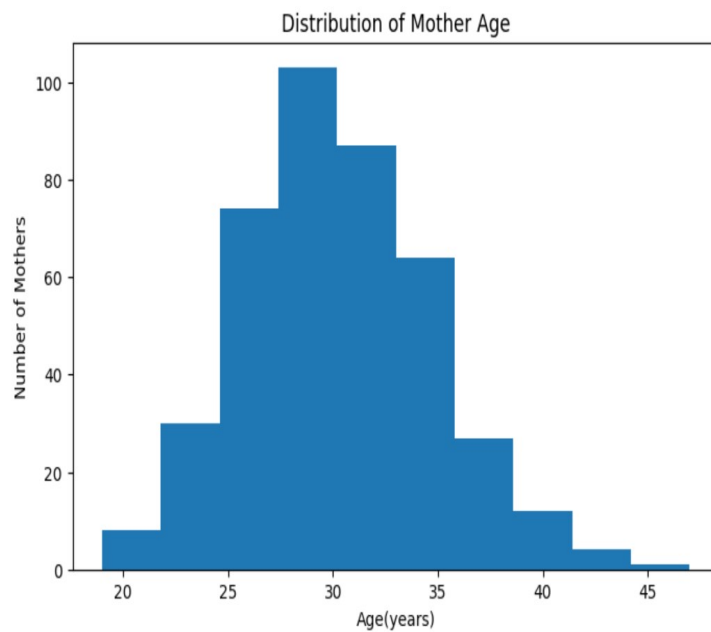
Key Findings

- HADS-A vs EPDS = 0.79 (Strong Positive Correlation)
- HADS Total vs EPDS = 0.72 (Strong Positive Correlation)
- CBTS Total vs EPDS = 0.82 (Strong Positive Correlation from heatmap)
- Education level was associated with lower EPDS, HADS and CBTS scores.
- Baby age showed negligible relationships with maternal mental health outcomes.
- Night awakenings showed weak positive relationships with EPDS and HADS.
- Sleep methods 2 and 5 had higher average EPDS scores than method 4.

Visual Analysis

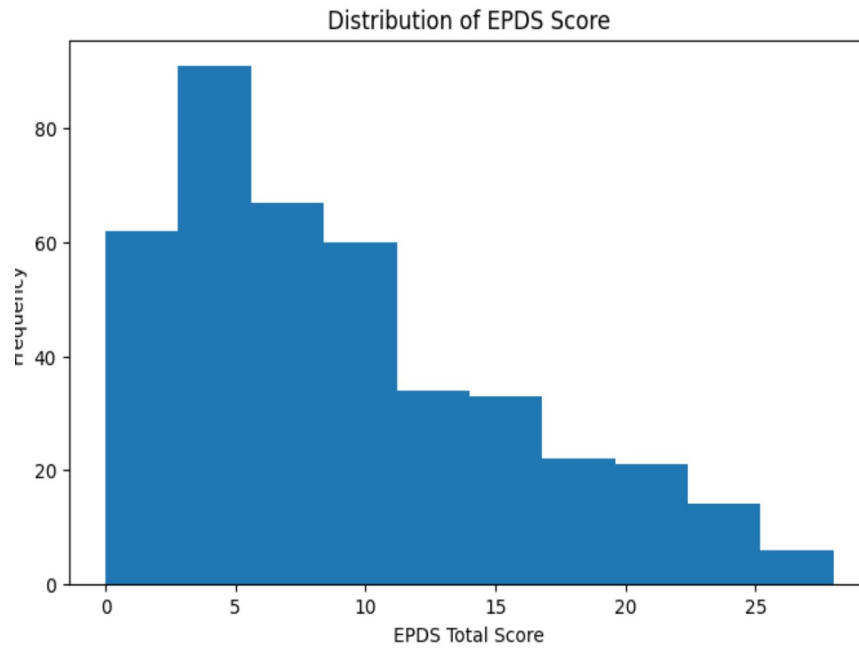


Correlation Heatmap Interpretation: Strong relationships exist among EPDS, HADS and CBTS. Age and night awakenings show weak associations.

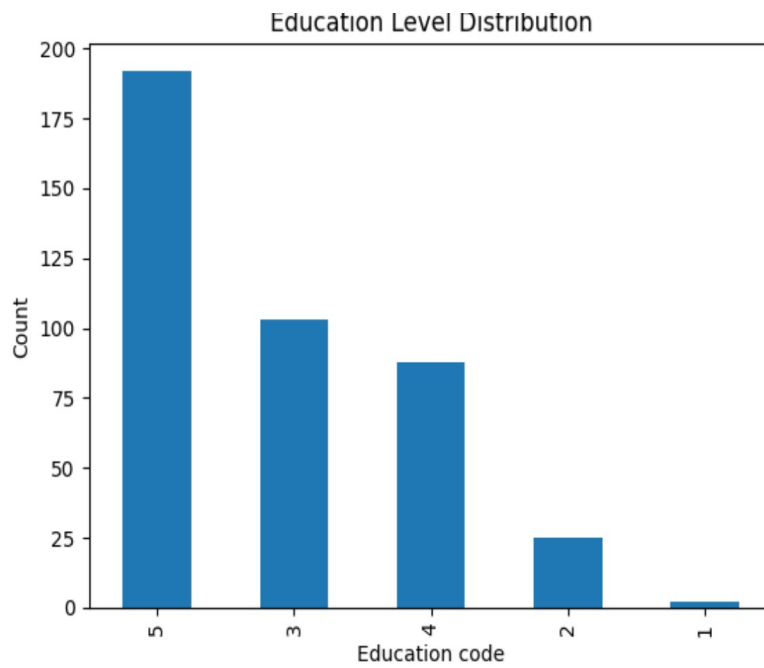


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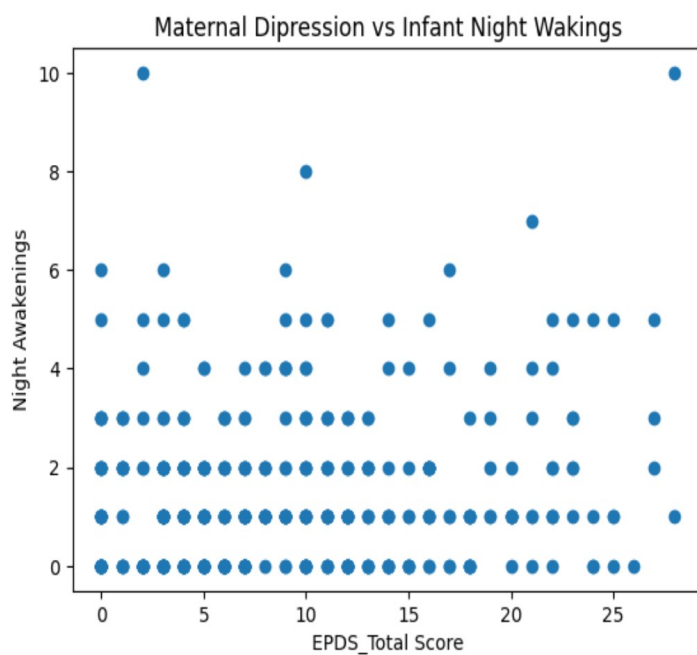
Mother Age Distribution: Most mothers are concentrated between 27 and 33 years of age.



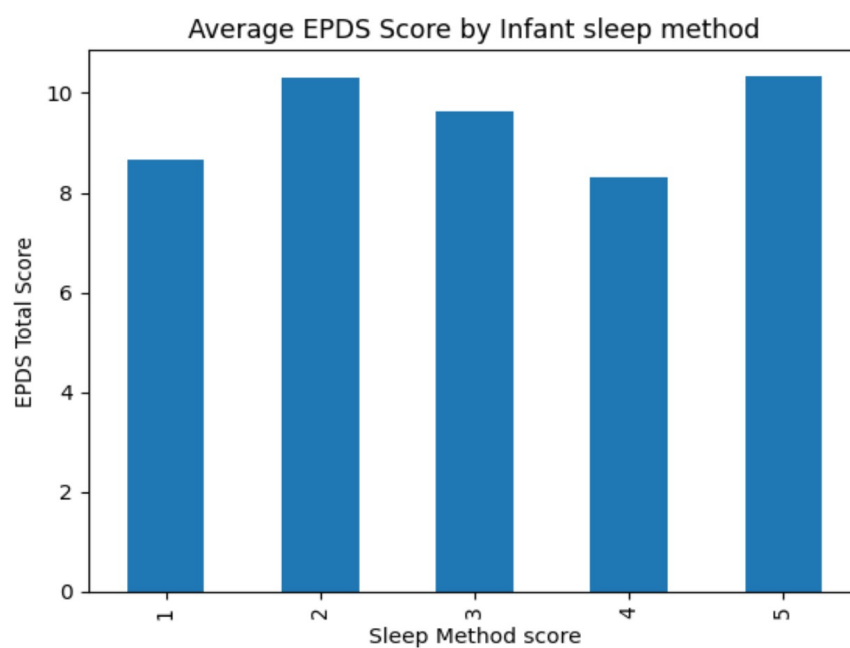
EPDS Distribution: Depression scores are concentrated in lower to moderate ranges.



Education Distribution: Most participants belong to higher education categories.



Maternal Depression vs Night Awakenings: Weak positive relationship.



Average EPDS by Infant Sleep Method: Sleep methods 2 and 5 show higher average EPDS scores.

Project Workflow

Dataset Collection → Data Cleaning → Missing Value Analysis → Feature Engineering (EPDS_Total, HADS_Total) → Exploratory Data Analysis → Correlation Analysis → Visualization → Findings & Conclusions

Conclusion

Maternal anxiety, depression and trauma measures were strongly interconnected. Education emerged as the strongest demographic factor associated with maternal mental health outcomes. Infant sleep characteristics and baby age demonstrated comparatively weaker relationships.

References & Appendix

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook. Dataset: Maternal Mental Health and Infant Sleep Dataset.